$$\lambda_1 \to \frac{1}{4}\left(g_1{}^2 + g_2{}^2\right) +$$
$$\frac{1}{16\pi^2}\Bigg(\frac{1}{24}\left(-3 g_1{}^4 - 5 g_2{}^4\right) + \frac{4}{9} C_{\phi 1^2} g_2{}^4 LF_{3,0}[m_2] + \frac{2}{3} C_{\phi 1^2} g_2{}^4 LF_{4,-1}[m_2] - \frac{32}{45} C_{\phi 1^2} g_2{}^4 LF_{5,-2}[m_2] -$$
$$\frac{1}{12}\sum_p g_1{}^4 LF_{2,0}[m_{\tilde{d}}{}^p] - \frac{1}{4}\sum_p g_1{}^4 LF_{2,0}[m_{\tilde{e}}{}^p] - \frac{1}{8}\sum_p \left(g_1{}^4 + g_2{}^4\right) LF_{2,0}[m_{\tilde{l}}{}^p] -$$
$$\frac{5}{18}\sum_p C_{\phi 1^2} g_2{}^4 LF_{3,0}[m_{\tilde{l}}{}^p] + \frac{1}{12}\sum_p C_{\phi 1^2} g_2{}^4 LF_{4,-1}[m_{\tilde{l}}{}^p] + \frac{8}{45}\sum_p C_{\phi 1^2} g_2{}^4 LF_{5,-2}[m_{\tilde{l}}{}^p] +$$
$$\left(-\frac{1}{2}\overline{y_u}^{pr} y_u{}^{pr}\left(g_1{}^2 - 3 g_2{}^2\right) - \frac{1}{24}\sum_p\left(g_1{}^4 + 9 g_2{}^4\right)\right) LF_{2,0}[m_{\tilde{q}}{}^p] +$$
$$\frac{1}{6} C_{\phi 1^2} g_2{}^2\left(36\,\overline{y_u}^{pr} y_u{}^{pr} - 5\sum_p g_2{}^2\right) LF_{3,0}[m_{\tilde{q}}{}^p] + \frac{1}{4} C_{\phi 1^2} g_2{}^2\left(-24\,\overline{y_u}^{pr} y_u{}^{pr} + \sum_p g_2{}^2\right) LF_{4,-1}[m_{\tilde{q}}{}^p] +$$
$$\frac{8}{15}\sum_p C_{\phi 1^2} g_2{}^4 LF_{5,-2}[m_{\tilde{q}}{}^p] - \frac{1}{3}\sum_p g_1{}^4 LF_{2,0}[m_{\tilde{u}}{}^p] + 2 g_1{}^2\,\overline{y_u}^{pr} y_u{}^{pr} LF_{2,0}[m_{\tilde{u}}{}^r] +$$
$$\frac{2}{9} C_{\phi 1^2} g_2{}^4 LF_{3,0}[\tilde{\mu}] + \frac{1}{3} C_{\phi 1^2} g_2{}^4 LF_{4,-1}[\tilde{\mu}] - \frac{16}{45} C_{\phi 1^2} g_2{}^4 LF_{5,-2}[\tilde{\mu}] + \frac{1}{2} g_1{}^4 LF_{2,2,-2}[m_1, \tilde{\mu}] +$$
$$\frac{1}{2} g_1{}^4 m_1{}^2 LF_{2,2,-1}[m_1, \tilde{\mu}] - \frac{8}{3} C_{\phi 1^2} g_2{}^4 LF_{2,1,0}[m_2, \tilde{\mu}] + \frac{5}{2} g_2{}^4 LF_{2,2,-2}[m_2, \tilde{\mu}] +$$
$$\frac{1}{6} g_2{}^4\left(-34 C_{\phi 1^2} + 3 m_2{}^2\right) LF_{2,2,-1}[m_2, \tilde{\mu}] + \frac{2}{3} m_2\,\tilde{\mu}\, g_2{}^4\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) LF_{2,2,0}[m_2, \tilde{\mu}] +$$
$$\frac{4}{3} C_{\phi 1^2} g_2{}^4 LF_{3,1,-1}[m_2, \tilde{\mu}] + \frac{28}{3} C_{\phi 1^2} g_2{}^4 LF_{3,2,-2}[m_2, \tilde{\mu}] -$$
$$\frac{1}{3} m_2 g_2{}^4\left(18 m_2 C_{\phi 1^2} + \tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\right) LF_{3,2,-1}[m_2, \tilde{\mu}] - 4 C_{\phi 1^2} g_2{}^4 LF_{3,3,-3}[m_2, \tilde{\mu}] +$$
$$4 C_{\phi 1^2} g_2{}^4 m_2{}^2 LF_{3,3,-2}[m_2, \tilde{\mu}] - 4 C_{\phi 1^2} g_2{}^4 LF_{4,2,-3}[m_2, \tilde{\mu}] + 4 C_{\phi 1^2} g_2{}^4 m_2{}^2 LF_{4,2,-2}[m_2, \tilde{\mu}] -$$
$$g_1{}^2\tilde{\mu}^2\,\overline{y_d}^{pr} y_d{}^{pr} LF_{2,1,0}[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p] - \frac{1}{2}\tilde{\mu}\, g_1{}^2\left(C_{\phi1\phi2}\, y_d{}^{pr}\,\overline{a_d}^{pr} + \overline{y_d}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_d{}^{pr}\right)\right)$$
$$LF_{2,2,0}[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p] - g_1{}^2\tilde{\mu}^2\,\overline{y_e}^{pr} y_e{}^{pr} LF_{2,1,0}[m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^p] -$$
$$\frac{1}{2}\tilde{\mu}\, g_1{}^2\left(C_{\phi1\phi2}\, y_e{}^{pr}\,\overline{a_e}^{pr} + \overline{y_e}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_e{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_e{}^{pr}\right)\right) LF_{2,2,0}[m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^p] +$$
$$g_1{}^2\tilde{\mu}^2\,\overline{y_e}^{pr} y_e{}^{pr} LF_{2,1,0}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r] - \frac{1}{2}\tilde{\mu}^2\,\overline{y_e}^{pr} y_e{}^{pr}\left(g_1{}^2 + g_2{}^2\right) LF_{3,1,-1}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r] +$$
$$\tilde{\mu}\, g_1{}^2\left(C_{\phi1\phi2}\, y_e{}^{pr}\,\overline{a_e}^{pr} + \overline{y_e}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_e{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_e{}^{pr}\right)\right) LF_{3,1,0}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r] +$$
$$\frac{1}{2}\tilde{\mu}\, g_1{}^2\left(C_{\phi1\phi2}\, y_e{}^{pr}\,\overline{a_e}^{pr} + \overline{y_e}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_e{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_e{}^{pr}\right)\right) LF_{3,2,-1}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r] -$$
$$\frac{3}{4}\tilde{\mu}\left(3 g_1{}^2 + g_2{}^2\right)\left(C_{\phi1\phi2}\, y_e{}^{pr}\,\overline{a_e}^{pr} + \overline{y_e}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_e{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_e{}^{pr}\right)\right) LF_{4,1,-1}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r] +$$
$$\frac{1}{3}\tilde{\mu}\left(\left(3 g_1{}^2 + 2 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr} a_e{}^{pr} + C_{\phi1\phi2}\, y_e{}^{pr}\,\overline{a_e}^{pr}\right) + 2 C_{\phi 1^2}\tilde{\mu}\,\overline{y_e}^{pr} y_e{}^{pr}\left(3 g_1{}^2 + g_2{}^2\right)\right)$$
$$LF_{5,1,-2}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r] + g_1{}^2\tilde{\mu}^2\,\overline{y_d}^{pr} y_d{}^{pr} LF_{2,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] - \frac{3}{2}\tilde{\mu}^2\,\overline{y_d}^{pr} y_d{}^{pr}\left(g_1{}^2 + g_2{}^2\right)$$
$$LF_{3,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] + \tilde{\mu}\, g_1{}^2\left(C_{\phi1\phi2}\, y_d{}^{pr}\,\overline{a_d}^{pr} + \overline{y_d}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_d{}^{pr}\right)\right) LF_{3,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] +$$
$$\frac{1}{2}\tilde{\mu}\, g_1{}^2\left(C_{\phi1\phi2}\, y_d{}^{pr}\,\overline{a_d}^{pr} + \overline{y_d}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_d{}^{pr}\right)\right) LF_{3,2,-1}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] -$$
$$\frac{3}{4}\tilde{\mu}\left(5 g_1{}^2 + 3 g_2{}^2\right)\left(C_{\phi1\phi2}\, y_d{}^{pr}\,\overline{a_d}^{pr} + \overline{y_d}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pr} + 2 C_{\phi 1^2}\tilde{\mu}\, y_d{}^{pr}\right)\right) LF_{4,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] +$$
$$\tilde{\mu}\left(\left(3 g_1{}^2 + 2 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr} a_d{}^{pr} + C_{\phi1\phi2}\, y_d{}^{pr}\,\overline{a_d}^{pr}\right) + 2 C_{\phi 1^2}\tilde{\mu}\,\overline{y_d}^{pr} y_d{}^{pr}\left(3 g_1{}^2 + g_2{}^2\right)\right)$$
$$LF_{5,1,-2}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] - 3\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{pt} y_u{}^{sr} LF_{1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] -$$
$$6 C_{\phi 1^2}\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{pt} y_u{}^{sr} LF_{2,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] + 6 C_{\phi 1^2}\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{pt} y_u{}^{sr} LF_{3,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] -$$
$$\frac{1}{2}\overline{a_u}^{pr} a_u{}^{pr}\left(g_1{}^2 - 3 g_2{}^2\right) LF_{2,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] +$$
$$\frac{1}{4}\left(-2 C_{\phi 1^2}\overline{a_u}^{pr} a_u{}^{pr}\left(g_1{}^2 - 7 g_2{}^2\right) - \tilde{\mu}\left(g_1{}^2 - 5 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr} a_u{}^{pr} + C_{\phi1\phi2}\, y_u{}^{pr}\,\overline{a_u}^{pr}\right)\right)$$
$$LF_{2,2,0}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] + \frac{3}{2} g_2{}^2\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr} a_u{}^{pr} + \overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{pr}\right)\right)$$
$$LF_{3,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] - g_2{}^2\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr} a_u{}^{pr} + \overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{pr}\right)\right) LF_{3,2,-1}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] -$$
$$\frac{3}{2} g_2{}^2\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr} a_u{}^{pr} + \overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{pr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{pr}\right)\right) LF_{4,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] +$$
$$\frac{1}{2}\overline{a_u}^{pr} a_u{}^{pr}\left(7 g_1{}^2 + 3 g_2{}^2\right) LF_{2,1,0}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p] - \frac{3}{2}\overline{a_u}^{pr} a_u{}^{pr}\left(g_1{}^2 + g_2{}^2\right) LF_{3,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p] +$$
$$\left(C_{\phi 1^2}\overline{a_u}^{pr} a_u{}^{pr}\left(7 g_1{}^2 + g_2{}^2\right) + \frac{1}{2}\tilde{\mu}\left(7 g_1{}^2 + 2 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr} a_u{}^{pr} + C_{\phi1\phi2}\, y_u{}^{pr}\,\overline{a_u}^{pr}\right)\right) LF_{3,1,0}[m_{\tilde{u}}{}^r,$$
$$m_{\tilde{q}}{}^p] + \frac{1}{4}\left(2 C_{\phi 1^2}\overline{a_u}^{pr} a_u{}^{pr}\left(g_1{}^2 - 7 g_2{}^2\right) + \tilde{\mu}\left(g_1{}^2 - 5 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr} a_u{}^{pr} + C_{\phi1\phi2}\, y_u{}^{pr}\,\overline{a_u}^{pr}\right)\right)$$
$$LF_{3,2,-1}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p] + \Bigg(-3 C_{\phi 1^2}\overline{a_u}^{pr} a_u{}^{pr}\left(5 g_1{}^2 + g_2{}^2\right) -$$
$$\frac{3}{2}\tilde{\mu}\left(5 g_1{}^2 + 2 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr} a_u{}^{pr} + C_{\phi1\phi2}\, y_u{}^{pr}\,\overline{a_u}^{pr}\right)\Bigg) LF_{4,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p] +$$
$$\left(2 C_{\phi 1^2}\overline{a_u}^{pr} a_u{}^{pr}\left(3 g_1{}^2 + g_2{}^2\right) + \tilde{\mu}\left(3 g_1{}^2 + 2 g_2{}^2\right)\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{pr} a_u{}^{pr} + C_{\phi1\phi2}\, y_u{}^{pr}\,\overline{a_u}^{pr}\right)\right)$$
$$LF_{5,1,-2}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p] - 3\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{pt} y_u{}^{sr} LF_{1,1,0}[m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t] -$$
$$\frac{3}{4} g_1{}^2\left(g_1{}^2 + g_2{}^2\right) LF_{2,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} g_1{}^2\left(g_1{}^2 + g_2{}^2\right) LF_{3,1,-2}[\tilde{\mu}, m_1] -$$
$$2 C_{\phi 1^2} g_1{}^2\left(g_1{}^2 + g_2{}^2\right) LF_{3,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} m_1\tilde{\mu}\, g_1{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\left(g_1{}^2 + g_2{}^2\right) LF_{3,1,0}[\tilde{\mu}, m_1] +$$
$$\frac{3}{2} C_{\phi 1^2} g_1{}^4 LF_{3,2,-2}[\tilde{\mu}, m_1] + \frac{1}{2} m_1 g_1{}^4\left(3 m_1 C_{\phi 1^2} - 2\tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\right) LF_{3,2,-1}[\tilde{\mu}, m_1] +$$
$$\frac{1}{3} C_{\phi 1^2} g_1{}^2\left(12 g_1{}^2 + 7 g_2{}^2\right) LF_{4,1,-2}[\tilde{\mu}, m_1] -$$
$$\frac{1}{4} m_1\tilde{\mu}\, g_1{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\left(6 g_1{}^2 + 5 g_2{}^2\right) LF_{4,1,-1}[\tilde{\mu}, m_1] - C_{\phi 1^2} g_1{}^4 LF_{4,2,-3}[\tilde{\mu}, m_1] +$$
$$m_1 g_1{}^4\left(-m_1 C_{\phi 1^2} + \tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\right) LF_{4,2,-2}[\tilde{\mu}, m_1] - \frac{2}{3} C_{\phi 1^2} g_1{}^2\left(3 g_1{}^2 + g_2{}^2\right) LF_{5,1,-3}[\tilde{\mu}, m_1] +$$
$$\frac{1}{3} m_1\tilde{\mu}\, g_1{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\left(3 g_1{}^2 + 2 g_2{}^2\right) LF_{5,1,-2}[\tilde{\mu}, m_1] - \frac{9}{4} g_2{}^2\left(g_1{}^2 + g_2{}^2\right) LF_{2,1,-1}[\tilde{\mu}, m_2] +$$
$$\frac{3}{2} g_2{}^2\left(g_1{}^2 + g_2{}^2\right) LF_{3,1,-2}[\tilde{\mu}, m_2] - \frac{2}{3} C_{\phi 1^2} g_2{}^2\left(9 g_1{}^2 + g_2{}^2\right) LF_{3,1,-1}[\tilde{\mu}, m_2] +$$
$$\frac{1}{6} m_2\tilde{\mu}\, g_2{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\left(9 g_1{}^2 + 5 g_2{}^2\right) LF_{3,1,0}[\tilde{\mu}, m_2] + \frac{85}{6} C_{\phi 1^2} g_2{}^4 LF_{3,2,-2}[\tilde{\mu}, m_2] +$$
$$\frac{1}{6} m_2 g_2{}^4\left(9 m_2 C_{\phi 1^2} - 28\tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\right) LF_{3,2,-1}[\tilde{\mu}, m_2] +$$
$$3 C_{\phi 1^2} g_2{}^2\left(4 g_1{}^2 + g_2{}^2\right) LF_{4,1,-2}[\tilde{\mu}, m_2] -$$
$$\frac{1}{4} m_2\tilde{\mu}\, g_2{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\left(18 g_1{}^2 + 11 g_2{}^2\right) LF_{4,1,-1}[\tilde{\mu}, m_2] - 7 C_{\phi 1^2} g_2{}^4 LF_{4,2,-3}[\tilde{\mu}, m_2] +$$
$$m_2 g_2{}^4\left(-m_2 C_{\phi 1^2} + 4\tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\right) LF_{4,2,-2}[\tilde{\mu}, m_2] - 2 C_{\phi 1^2} g_2{}^2\left(3 g_1{}^2 + g_2{}^2\right) LF_{5,1,-3}[\tilde{\mu}, m_2] +$$
$$m_2\tilde{\mu}\, g_2{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\left(3 g_1{}^2 + 2 g_2{}^2\right) LF_{5,1,-2}[\tilde{\mu}, m_2] + 8 C_{\phi 1^2} g_1{}^2 g_2{}^2 LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] +$$
$$\frac{1}{2} g_1{}^2 g_2{}^2\left(12 m_1 m_2 C_{\phi 1^2} - \tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)(7 m_1 + 5 m_2)\right) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] -$$
$$4 C_{\phi 1^2} g_1{}^2 g_2{}^2 LF_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] +$$
$$2 g_1{}^2 g_2{}^2\left(-2 m_1 m_2 C_{\phi 1^2} + \tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)(m_1 + m_2)\right) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] +$$
$$6 C_{\phi 1^2}\tilde{\mu}^2\,\overline{y_d}^{pr} y_d{}^{sr}\,\overline{y_u}^{st} y_u{}^{pt} LF_{2,2,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r] - 6\,\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr} a_u{}^{sr}$$
$$LF_{1,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r] - \frac{3}{2}\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{2,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r] +$$
$$\frac{3}{2}\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{3,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r] -$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{2,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{3,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{4}\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{sr} + 3 C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{2,2,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^s] -$$
$$6\,\overline{y_u}^{st} y_u{}^{sr}\,\overline{a_u}^{pr} a_u{}^{pt} LF_{1,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t] +$$
$$\frac{9}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{2,2,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^s] +$$
$$\frac{3}{2}\tilde{\mu}\left(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr}\,\overline{y_u}^{st} y_u{}^{pt} a_d{}^{sr} + C_{\phi1\phi2}\, y_d{}^{pr}\,\overline{y_u}^{pt} y_u{}^{st}\,\overline{a_d}^{sr}\right) LF_{2,2,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r] -$$
$$\frac{3}{2}\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{2,1,1,0}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] +$$
$$\frac{3}{2}\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{3,1,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] -$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{2,1,1,0}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{2}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{3,1,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{4}\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(4 C_{\phi 1^2} a_u{}^{sr} - C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{2,2,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p] -$$
$$\frac{3}{4}\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{2,2,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p] -$$
$$3\,\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(2 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{2,1,1,0}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] +$$
$$3\,\overline{y_u}^{st} y_u{}^{pt}\,\overline{a_u}^{pr}\left(2 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{3,1,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] -$$
$$6\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt}\left(C_{\phi 1^2}\overline{a_u}^{pr} + \overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\right) LF_{2,1,1,0}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t] +$$
$$6\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt}\left(C_{\phi 1^2}\overline{a_u}^{pr} + \overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\right) LF_{3,1,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t] +$$
$$3\,\overline{y_u}^{st} y_u{}^{sr}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr} a_u{}^{pt} + \overline{a_u}^{pr}\left(2 C_{\phi 1^2} a_u{}^{pt} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{pt}\right)\right) LF_{2,2,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p] -$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{2,1,1,0}[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] +$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u{}^{sr} a_u{}^{pt} LF_{3,1,1,-1}[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] -$$
$$6\,\overline{y_u}^{st} y_u{}^{sr}\,\overline{a_u}^{pr}\left(C_{\phi 1^2} a_u{}^{pt} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{pt}\right) LF_{2,1,1,0}[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] +$$
$$6\,\overline{y_u}^{st} y_u{}^{sr}\,\overline{a_u}^{pr}\left(C_{\phi 1^2} a_u{}^{pt} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{pt}\right) LF_{3,1,1,-1}[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] +$$
$$g_1{}^2 g_2{}^2 LF_{2,1,1,-2}[\tilde{\mu}, m_1, m_2] + g_1{}^2 g_2{}^2\left(-3 C_{\phi 1^2} + m_1 m_2\right) LF_{2,1,1,-1}[\tilde{\mu}, m_1, m_2] +$$
$$m_1\tilde{\mu}\, g_1{}^2 g_2{}^2\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right) LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + 7 C_{\phi 1^2} g_1{}^2 g_2{}^2 LF_{3,1,1,-2}[\tilde{\mu}, m_1, m_2] +$$
$$\frac{1}{4} g_1{}^2 g_2{}^2\left(12 m_1 m_2 C_{\phi 1^2} - \tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)(9 m_1 + 5 m_2)\right) LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] -$$
$$4 C_{\phi 1^2} g_1{}^2 g_2{}^2 LF_{4,1,1,-3}[\tilde{\mu}, m_1, m_2] + \frac{1}{2} g_1{}^2 g_2{}^2\left(-4 m_1 m_2 C_{\phi 1^2} + 3\tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)(m_1 + m_2)\right)$$
$$LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - 4 C_{\phi 1^2} g_1{}^2 g_2{}^2 LF_{3,2,1,-3}[\tilde{\mu}, m_2, m_1] +$$
$$2 g_1{}^2 g_2{}^2\left(-2 m_1 m_2 C_{\phi 1^2} + \tilde{\mu}\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)(m_1 + m_2)\right) LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] -$$
$$3\tilde{\mu}^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st} y_d{}^{pt} y_d{}^{sr} LF_{1,1,1,1,0}[m_{\tilde{d}}{}^r, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] +$$
$$\frac{3}{2}\tilde{\mu}^3\,\overline{y_d}^{st} y_d{}^{sr}\left(\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr} a_d{}^{pt} - C_{\phi1\phi2}\, y_d{}^{pt}\,\overline{a_d}^{pr}\right) LF_{2,2,1,1,-1}[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s] +$$
$$\frac{3}{2}\tilde{\mu}^3\,\overline{y_d}^{st} y_d{}^{sr}\left(-\overline{C_{\phi1\phi2}}\,\overline{y_d}^{pr} a_d{}^{pt} + C_{\phi1\phi2}\, y_d{}^{pt}\,\overline{a_d}^{pr}\right) LF_{2,2,1,1,-1}[m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^s] -$$
$$\tilde{\mu}^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st} y_e{}^{pt} y_e{}^{sr} LF_{1,1,1,1,0}[m_{\tilde{e}}{}^r, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^p, m_{\tilde{l}}{}^s] +$$
$$\frac{1}{2}\tilde{\mu}^3\,\overline{y_e}^{st} y_e{}^{sr}\left(\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr} a_e{}^{pt} - C_{\phi1\phi2}\, y_e{}^{pt}\,\overline{a_e}^{pr}\right) LF_{2,2,1,1,-1}[m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^s] +$$
$$\frac{1}{2}\tilde{\mu}^3\,\overline{y_e}^{st} y_e{}^{sr}\left(-\overline{C_{\phi1\phi2}}\,\overline{y_e}^{pr} a_e{}^{pt} + C_{\phi1\phi2}\, y_e{}^{pt}\,\overline{a_e}^{pr}\right) LF_{2,2,1,1,-1}[m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^s] -$$
$$\tilde{\mu}^3\,\overline{y_e}^{st} y_e{}^{sr}\left(C_{\phi1\phi2}\, y_e{}^{pt}\,\overline{a_e}^{pr} + \overline{y_e}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_e{}^{pt} + 2 C_{\phi 1^2}\tilde{\mu}\, y_e{}^{pt}\right)\right) LF_{2,1,1,1,0}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^s] +$$
$$\tilde{\mu}^3\,\overline{y_e}^{st} y_e{}^{sr}\left(C_{\phi1\phi2}\, y_e{}^{pt}\,\overline{a_e}^{pr} + \overline{y_e}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_e{}^{pt} + 2 C_{\phi 1^2}\tilde{\mu}\, y_e{}^{pt}\right)\right) LF_{3,1,1,1,-1}[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^s] -$$
$$3\tilde{\mu}^3\,\overline{y_d}^{st} y_d{}^{sr}\left(C_{\phi1\phi2}\, y_d{}^{pt}\,\overline{a_d}^{pr} + \overline{y_d}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pt} + 2 C_{\phi 1^2}\tilde{\mu}\, y_d{}^{pt}\right)\right) LF_{2,1,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s] +$$
$$3\tilde{\mu}^3\,\overline{y_d}^{st} y_d{}^{sr}\left(C_{\phi1\phi2}\, y_d{}^{pt}\,\overline{a_d}^{pr} + \overline{y_d}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pt} + 2 C_{\phi 1^2}\tilde{\mu}\, y_d{}^{pt}\right)\right) LF_{3,1,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, m_{\tilde{d}}{}^t,$$
$$m_{\tilde{q}}{}^s] + \frac{3}{2}\tilde{\mu}\, y_d{}^{sr} a_u{}^{pt}\left(C_{\phi1\phi2}\,\overline{a_d}^{pr}\,\overline{a_u}^{st} + \overline{C_{\phi1\phi2}}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_u}^{st}\right) LF_{2,2,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}\,\overline{a_u}^{pr}\left(\overline{C_{\phi1\phi2}}\, a_d{}^{pt} a_u{}^{sr} + \tilde{\mu}\, y_d{}^{pt}\left(4 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right)\right)$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^t, m_{\tilde{u}}{}^r] - 3\,\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u{}^{pt} a_u{}^{sr} LF_{1,1,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t] -$$
$$\frac{3}{2}\left(C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{st}\,\overline{a_u}^{pt}\,\overline{a_u}^{sr} a_u{}^{pr} + \overline{a_u}^{pr} a_u{}^{pt} a_u{}^{sr}\left(4 C_{\phi 1^2}\overline{a_u}^{st} + \overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}\right)\right)$$
$$LF_{2,1,1,1,0}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t] + 6 C_{\phi 1^2}\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u{}^{pt} a_u{}^{sr} LF_{3,1,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{2}\tilde{\mu}\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}\,\overline{a_u}^{pr} a_u{}^{pt} a_u{}^{sr} + C_{\phi1\phi2}\, y_u{}^{st}\,\overline{a_u}^{pt}\,\overline{a_u}^{sr} a_u{}^{pr}\right) LF_{3,1,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t, m_{\tilde{u}}{}^r] +$$
$$\frac{3}{4}\overline{a_u}^{pr} a_u{}^{pt}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st} a_u{}^{sr} + \overline{a_u}^{st}\left(4 C_{\phi 1^2} a_u{}^{sr} + 3 C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right)\right)$$
$$LF_{2,2,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t] + \frac{3}{4}\overline{a_u}^{pr} a_u{}^{pt}$$
$$\left(3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st} a_u{}^{sr} + \overline{a_u}^{st}\left(4 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right)\right) LF_{2,2,1,1,-1}[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r] -$$
$$\frac{3}{2}\tilde{\mu}\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}\,\overline{a_u}^{pr} a_u{}^{pt} a_u{}^{sr} + C_{\phi1\phi2}\, y_u{}^{st}\,\overline{a_u}^{pt}\,\overline{a_u}^{sr} a_u{}^{pr}\right) LF_{2,1,1,1,0}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{2}\tilde{\mu}\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st}\,\overline{a_u}^{pr} a_u{}^{pt} a_u{}^{sr} + C_{\phi1\phi2}\, y_u{}^{st}\,\overline{a_u}^{pt}\,\overline{a_u}^{sr} a_u{}^{pr}\right) LF_{3,1,1,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t, m_{\tilde{u}}{}^r] +$$
$$\frac{3}{4}\tilde{\mu}\,\overline{a_u}^{pr} a_u{}^{pt}\left(\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st} a_u{}^{sr} - C_{\phi1\phi2}\, y_u{}^{sr}\,\overline{a_u}^{st}\right) LF_{2,2,1,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{4}\tilde{\mu}\,\overline{a_u}^{pr} a_u{}^{pt}\left(-\overline{C_{\phi1\phi2}}\,\overline{y_u}^{st} a_u{}^{sr} + C_{\phi1\phi2}\, y_u{}^{sr}\,\overline{a_u}^{st}\right) LF_{2,2,1,1,-1}[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r] -$$
$$3\,\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u{}^{pt}\left(2 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{2,1,1,1,0}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t] +$$
$$3\,\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u{}^{pt}\left(2 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right) LF_{3,1,1,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t] +$$
$$\frac{3}{2}\overline{a_u}^{pr} a_u{}^{pt}\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st} a_u{}^{sr} + \overline{a_u}^{st}\left(2 C_{\phi 1^2} a_u{}^{sr} + C_{\phi1\phi2}\tilde{\mu}\, y_u{}^{sr}\right)\right)$$
$$LF_{2,2,1,1,-1}[m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s] - 3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}\,\overline{a_u}^{pr} a_u{}^{pt} a_u{}^{sr} LF_{2,1,1,1,0}[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r] +$$
$$3\,\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_u}^{st}\,\overline{a_u}^{pr} a_u{}^{pt} a_u{}^{sr} LF_{3,1,1,1,-1}[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r]\Bigg)$$